

21 DAYS POWER BI DASHBOARD CHALLENGE

OBJECTIVES OF CHALLENGE

- Learn how to connect and clean live datasets in Power BI.
 - Understand how to model data with relationships and schema.
 - Practice writing basic to advance DAX formulas for analysis and insights.
 - Gain skills to design interactive and visually appealing dashboards.
 - Learn how to document, publish, and present a complete Power BI project.
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LIVE DATASETS:

- Datagov.in- <https://www.data.gov.in/catalogs>
 - Hubwise Data Hubà https://hub.huwise.com/explore/?orderBy=updated_at+DESC
 - Open City Urban Data Portalà <https://data.opencity.in/dataset>
 - <https://mavenanalytics.io/data-playground>
 - <https://www.sec.gov/data-research/sec-markets-data/financial-statement-data-sets>
 - **BSE SENSEX Stock Market Index**
 - **National Commodity & Derivatives Exchange Limited (NCDEX)**
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TASK 1 – DATASET SELECTION & IMPORT

- Choose one dataset from the given list.
 - Import data into Power BI Desktop (via Web).
 -  *Task:* Verify data is loaded correctly.
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TASK 2 – DATA CLEANING (POWER QUERY)

- Detect and remove duplicates or missing values.
- Standardize column names and data types.

- Perform necessary transformations (split, merge, replace values).
 -  Task: Prepare a clean dataset for modelling.
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TASK 3 – DATA MODELING

- Identify Fact table(s) and Dimension table(s).
 - Build relationships (one-to-many / many-to-one).
 - Check cardinality and cross-filter direction.
 -  Task: Create a clear and correct data model.
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TASK 4 – BASIC DAX MEASURES

- Create aggregation measures such as:
 - **SUM** → total of a numeric column
 - **AVERAGE** → mean value of a numeric column
 - **COUNT / COUNTROWS** → number of records
 - **MIN / MAX** → smallest and largest values
 -  Task: Use these measures in simple visuals (Cards, Tables, Charts).
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TASK 5 – ADVANCED DAX

- Create measures using **time intelligence functions**.
 - Create measures for **comparisons across time periods**.
 - Create measures to calculate **growth or percentage change**.
 - Create measures to perform **ranking and top N analysis**.
 -  Task: Develop at least 3–5 advanced measures and apply them in visuals.
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TASK 6 – DASHBOARD VISUALIZATION (PART 1)

- Create visuals using:

- Bar, Column, Pie, Donut
 - Line, Area, Scatter
 - KPI & Gauge Cards
 -  *Task:* Show overall key metrics and trends.
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TASK 7 – DASHBOARD VISUALIZATION (PART 2)

- Add visuals:
 - Table & Matrix
 - Map, Tree Map, Combo Chart
 - Apply **conditional formatting** and **data bars**.
 - Create **hierarchies** (e.g., Year → Quarter → Month).
 -  *Task:* Build detailed visuals with drill-down capability.
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TASK 8 – DASHBOARD VISUALIZATION (PART 3)

- Enhance dashboard with:
 - Filters & Slicers
 - Drill-downs
 - Bookmarks & Buttons for navigation
 -  *Task:* Make the dashboard interactive and user-friendly.
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TASK 9 – FINAL DASHBOARD DESIGN & PUBLISHING

- Apply **color themes, alignment, and storytelling flow**.
 - Add **titles, legends, and tooltips** for clarity.
 - Publish the dashboard to **GitHub / shared drive**.
 -  *Task:* Produce a professional-quality dashboard.
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TASK 10 – DOCUMENTATION & SUBMISSION

- Document your project (Word/PDF) including:
 1. About the dataset chosen
 2. Data cleaning steps
 3. Data model (with screenshot)
 4. DAX measures created
 5. Insights discovered
- Submit final deliverables:
 - Power BI file (.pbix)
 - Exported dashboard (.pdf)
 - Documentation/ReadMe file uploaded to GitHub

“Master Power BI, tell powerful data stories, and level up as a confident data analyst! ”

POWER BI BASICS & ECOSYSTEM

Q1. Which of the following is **NOT** a part of the Power BI ecosystem?

- A) Power BI Service
 - B) Power BI Desktop
 - C) Power BI Compiler**
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Q2. What is the **primary function of Power BI Service** (PowerBI.com)?

- A) Data modeling and creating relationships
 - B) Data visualization, collaboration, and cloud hosting of reports**
 - C) Data extraction and transformation
 - D) Installing custom connectors for SQL Server
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Q3. Which of the following scenarios is **best suited for Power BI Service** instead of Power BI Report Server?

- A) A government agency with strict on-premises data regulations
 - B) A multinational company requiring cloud collaboration across regions**
 - C) A hospital needing local-only report hosting for compliance reasons
 - D) An organization with zero internet access
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Q4. In the Power BI ecosystem, **Power BI Desktop** is mainly used for:

- A) Hosting dashboards for end-users
 - B) Creating and designing reports, data modeling, and DAX calculations**
 - C) Real-time collaboration and publishing dashboards
 - D) Automating refresh schedules for reports
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Q5. In the Power BI ecosystem, where are **workspaces** primarily managed?

- A) Power BI Desktop
 - B) Power BI Service**
 - C) Power BI Gateway
 - D) Power Query Editor
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Q6. Which of the following is **NOT one of the three main views** in Power BI Desktop (left navigation pane)?

- A) Report View
 - B) Data View
 - C) Model View
 - D) Service View**
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Q7. In Power BI Desktop, the **Fields pane** is primarily used for:

- A) Managing applied steps in data transformation
 - B) Adding fields to visuals, measures, and hierarchies**
 - C) Controlling refresh schedules for datasets
 - D) Defining workspace permissions
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Q8. In **Model View**, you can:

- A) Change visual formatting for charts
 - B) Define relationships between tables**
 - C) Apply data type changes at source level
 - D) Create dashboards for publishing
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Q9. Which feature in the Power BI **Report View** enables drilling down into hierarchical data (e.g., Year → Quarter → Month)?

- A) Bookmarks
- B) Drill-through**

C) Hierarchies

D) Filters pane

Q10. The **Visualizations pane** in Power BI allows you to:

A) Create custom queries in M

B) Select chart types and customize formatting

C) Connect to SQL Server and fetch tables

D) Configure scheduled refresh in Service

POWER QUERY – DATA PREPARATION & TRANSFORMATION

Q1. When connecting to a JSON API, which step in **Power Query** is essential to flatten nested structures into a table?

- A) Create relationships in Model View
 - B) Use “To Table” and “Expand” transformations**
 - C) Write DAX calculated tables
 - D) Use DirectQuery mode
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Q2. Which Power BI feature allows connecting to multiple files (e.g., monthly sales CSVs) in a folder and automatically combining them into one dataset?

- A) Append Queries in Power Query
 - B) Combine Files (Folder Connector)**
 - C) Dataflows in Power BI Service
 - D) Relationships in Model View
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Q3. Power Query in Power BI is mainly used for:

- A) Data modeling and DAX calculations
 - B) Data cleaning, transformation, and shaping before loading into the model**
 - C) Real-time data visualization
 - D) Scheduling refreshes in Power BI Service
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Q4. Power Query transformations are written in which language?

- A) SQL
- B) M (Mashup Language)**
- C) DAX
- D) C#

Q5. In Power Query, the **Applied Steps pane** records:

- A) Visual formatting changes in Report View
 - B) Each transformation step applied to the dataset**
 - C) Relationships between tables
 - D) Scheduled refresh frequency
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Q6. Which transformation in Power Query allows creating **aggregations (sum, avg, count, etc.)**?

- A) Append Queries
 - B) Group By**
 - C) Pivot Columns
 - D) Replace Values
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Q7. When importing a CSV file into Power Query, Power BI automatically:

- A) Creates relationships between columns
 - B) Detects data types for each column**
 - C) Generates DAX measures
 - D) Removes null values
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Q8. Which option should you choose if you want to **clean data before loading it into the Power BI model**?

- A) Load
 - B) Transform Data**
 - C) Report View
 - D) DirectQuery
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Q9. If you import multiple monthly Excel files from a folder into Power Query, which option

consolidates them automatically?

- A) Merge Queries
 - B) Append Queries
 - C) Combine Files (Folder Connector)**
 - D) Create Relationships
-

Q10. Which of the following features is used to stack **multiple tables vertically** (same columns, same schema) in Power Query?

- A) Merge Queries
 - B) Append Queries**
 - C) Group By
 - D) Pivot Columns
-

Q12. Which option in Power Query is best suited for **joining two tables by a common key** (like CustomerID)?

- A) Append Queries
 - B) Merge Queries**
 - C) Group By
 - D) Unpivot Columns
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Q13. When importing an Excel sheet, if Power BI incorrectly detects the header row, what should you use?

- A) Promote Headers**
 - B) Use First Column as Headers
 - C) Replace Values
 - D) Split Column
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Q14. Which of the following is **TRUE** about the “Close & Load” options in Power Query?

- A) “Close & Load” loads the data directly into Excel only.

- B) **“Close & Load To” allows loading into Power BI model or as a connection only.**
 - C) “Close & Apply” is the same as “Close & Load.”
 - D) They permanently alter the source data file.
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Q15. In Power Query, which transformation is used to replace all null values in a column with a fixed value?

- A) Replace Errors
 - B) **Replace Values**
 - C) Fill Down
 - D) Fill Up
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Q16. “Fill Down” in Power Query works best when:

- A) You want to aggregate null rows.
 - B) The first row contains missing data.
 - C) **You need to replace nulls with the value directly above.**
 - D) You want to remove duplicate rows.
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Q17. Which transformation **cannot directly remove nulls** in Power Query?

- A) Remove Blank Rows
 - B) Replace Values with Zero
 - C) Remove Empty
 - D) **Remove Duplicates**
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Q18. When you apply **Remove Duplicates** in Power Query without selecting specific columns, what happens?

- A) **It removes rows only if all column values match.**
- B) It removes rows with at least one duplicate value in any column.
- C) It sorts the dataset automatically before removal.
- D) It deletes null values as well.

Q19. To keep only the **latest record per customer ID** in a dataset with duplicates, you should:

- A) Remove Duplicates on Customer ID directly.
 - B) Sort by Date descending, then Remove Duplicates on Customer ID.**
 - C) Group By Customer ID and count rows.
 - D) Append Queries and remove nulls.
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Q20. Which is true about sorting in Power Query?

- A) Sorting is permanent in the data model.
 - B) Sorting only affects display order in Query Editor.**
 - C) Sorting must be re-applied after refresh.
 - D) Sorting is applied at visualization level only.
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Q21. In Power Query, “Group By” followed by selecting **All Rows** will:

- A) Return unique groups only.
 - B) Collapse groups into aggregated values.
 - C) Nest original tables within each group.**
 - D) Delete duplicates automatically.
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Q22. Which aggregation function is **not available** in Power Query Group By?

- A) Sum
 - B) Average
 - C) Median
 - D) Mode**
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Q23. When you Group By a column in Power Query without applying aggregation:

- A) The data collapses into a unique set of values for that column.**
- B) It creates duplicate rows.

- C) It automatically applies Sum.
 - D) It deletes unrelated rows.
-

Q24. In Power Query, the difference between **Append Queries** and **Merge Queries** is:

- A) Append → Combines rows; Merge → Combines columns**
 - B) Append → Combines columns; Merge → Combines rows
 - C) Both are the same
 - D) Append is only for Excel sources
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Q25. Which feature in Power Query lets you **reorder transformation steps** if applied in the wrong order?

- A) Query Dependencies
 - B) Applied Steps Pane**
 - C) Data Types
 - D) Advanced Editor
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DATA MODELING & RELATIONSHIPS

Q1. Which of the following is the **primary goal** of data modeling in Power BI?

- A) To improve visual aesthetics of dashboards
 - B) To optimize storage size only
 - C) To create relationships between tables for accurate analysis**
 - D) To clean raw data sources before load
-

Q2. In Power BI, a **good data model** should follow:

- A) Flat tables with all data merged
 - B) Snowflake schema with many intermediate joins
 - C) Star schema with fact and dimension separation**
 - D) Denormalization of fact tables only
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Q3. Which is a disadvantage of **highly normalized models** in Power BI?

- A) Reduce redundancy
 - B) Easy to manage dimension tables
 - C) Require many joins → slower query performance**
 - D) Improve data consistency
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Q4. Denormalization in Power BI is preferred when:

- A) Data redundancy must be avoided at all costs
 - B) Query performance and simplicity matter more than strict consistency**
 - C) Data sources are OLTP systems
 - D) Data is very small in size
-

Q5. A **snowflake schema** is considered less efficient in Power BI because:

- A) It stores facts in multiple tables
 - B) It increases relationship complexity and slows performance**
 - C) It does not support DAX measures
 - D) It is only valid in OLTP systems
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Q6. Which of the following is **not a valid relationship type** in Power BI?

- A) One-to-One
 - B) One-to-Many
 - C) Many-to-Many
 - D) Many-to-One**
-

Q7. In Power BI, the **default cross-filter direction** for One-to-Many relationships is:

- A) Single (from dimension to fact)**
 - B) Single (from fact to dimension)
 - C) Both directions
 - D) No filtering by default
-

Q8. If you have **two fact tables** linked to the same dimension table, what issue may occur?

- A) Circular dependency
 - B) Ambiguous relationship paths**
 - C) Invalid DAX measure creation
 - D) Duplicate dimension entries
-

Q9. Which Power BI feature allows handling multiple relationships between the same two tables?

- A) Composite models
- B) Active/Inactive relationships**
- C) Role-playing dimensions

D) Cross-filter direction

Q10. Cardinality in Power BI relationships refers to:

- A) The direction of filters between tables
 - B) The uniqueness of values in the key column**
 - C) The data type compatibility of relationships
 - D) The number of relationships in the model
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Q11. Which of the following is a valid scenario for a **One-to-One relationship**?

- A) Each customer can place multiple orders
 - B) Each employee has one ID card, and each ID card belongs to one employee**
 - C) Each product belongs to many categories
 - D) Each sale transaction contains multiple products
-

Q12. A Many-to-Many relationship in Power BI can cause incorrect results unless:

- A) A bridge (intermediate) table is introduced**
 - B) It is converted to One-to-Many
 - C) The fact table is normalized
 - D) The relationship is disabled
-

Q13. Setting cross-filter direction to **Both** is risky because:

- A) It increases redundancy
 - B) It may create ambiguous filter paths leading to wrong results**
 - C) It prevents relationship activation
 - D) It disables DAX measures
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Q14. Cross-filter direction **Single** is most commonly applied from:

- A) Fact → Dimension

B) Dimension → Fact

C) Dimension ↔ Dimension

D) Fact ↔ Fact

Q15. When should you use cross-filter direction = **Both**?

A) When connecting two fact tables via a shared dimension

B) When handling role-playing dimensions

C) When performance must be improved

D) Always, since it's best practice

Q16. Which of the following is **NOT an advantage** of the star schema in Power BI?

A) Simpler relationships

B) Better performance for DAX

C) Reduces data redundancy

D) Easy to understand by business users

Q17. Which scenario **requires a bridge table** in a star schema?

A) Many-to-Many between Students and Courses

B) One-to-Many between Orders and Customers

C) One-to-One between Employees and ID Cards

D) Many-to-One between Sales and Products

Q18. Which of these is considered a **best practice for Power BI modeling**?

A) Use snowflake schema for better normalization

B) Create measures inside fact tables instead of dimensions

C) Avoid surrogate keys in dimensions

D) Use multiple active relationships for flexibility

DAX (DATA ANALYSIS EXPRESSIONS)

Q1. What does DAX stand for in Power BI?

- A) Data Analysis XML
 - B) Data Analysis Expressions**
 - C) Dynamic Analytics Expressions
 - D) Data Aggregation Expressions
-

Q2. DAX is primarily used for:

- A) Data Loading
 - B) Data Cleaning
 - C) Data Modeling and Calculations**
 - D) Report Publishing
-

Q3. A calculated column in DAX is evaluated:

- A) At query time
 - B) At data refresh/load time**
 - C) Only when exporting data
 - D) Only when the user applies filters
-

Q4. Which DAX function creates a calculated table?

- A) SUMMARIZE
 - B) CALCULATETABLE
 - C) ADDCOLUMNS
 - D) All of the above**
-

Q5. Unlike calculated columns, measures are:

- A) Stored in memory
 - B) Evaluated dynamically at query time**
 - C) Created only once per model
 - D) Cannot use aggregations
-

Q6. Which of the following is an example of a measure?

- A) Sales[Price] * Sales[Quantity]
 - B) SUM(Sales[Revenue])**
 - C) LEFT(Customer[Name], 3)
 - D) Sales[Discount] + 10
-

Q7. Which DAX operator is used for string concatenation?

- A) +
 - B) &**
 - C) ||
 - D) CONCAT()
-

Q8. The function IF(Sales[Amount] > 1000, "High", "Low") is an example of:

- A) Statistical function
 - B) Logical function**
 - C) Text function
 - D) Aggregation
-

Q9. Which function returns the average of a column?

- A) SUM
- B) DIVIDE
- C) AVERAGE**
- D) MEDIAN

Q10. Which text function extracts the first n characters from a string?

- A) RIGHT
 - B) MID
 - C) LEFT**
 - D) SUBSTRING
-

Q11. SUMX differs from SUM because:

- A) It aggregates over an expression row by row**
 - B) It is faster
 - C) It only works with integers
 - D) It ignores blank values
-

Q12. Which DAX function is used to create conditional aggregations?

- A) CALCULATE**
 - B) SUMX
 - C) FILTER
 - D) VALUES
-

Q13. What does DISTINCTCOUNT(Column) return?

- A) Number of rows
 - B) Unique values count**
 - C) Distinct rows in a table
 - D) Number of duplicates
-

Q14. DIVIDE([Total Sales], [Total Orders], 0) does what?

- A) Returns error on divide by zero
- B) Returns blank if denominator is zero**

C) Returns 0 if denominator is zero

D) Always rounds result

Q15. Which function returns the last date in the current filter context?

A) ENDOFMONTH

B) LASTDATE

C) MAX

D) CLOSINGBALANCEMONTH

Q16. DATESYTD(Date[Date]) returns:

A) All dates from start of current year to last date in filter

B) All dates from today to end of year

C) Only current year's last date

D) Year-over-year difference

Q17. Which function is used for previous period comparison?

A) PREVIOUSMONTH

B) PREVIOUSDAY

C) SAMEPERIODLASTYEAR

D) All of the above

Q18. Which function calculates a moving average in DAX?

A) AVERAGEX with DATESINPERIOD

B) CALCULATETABLE with VALUES

C) SUMX with EARLIER

D) RANKX with FILTER

Q19. Which function assigns a rank to rows based on expression?

- A) RANK
 - B) RANKX**
 - C) TOPN
 - D) ORDERBY
-

Q20. In RANKX, what does the order parameter control?

- A) Ascending/Descending ranking**
 - B) Filter context
 - C) Table join direction
 - D) Aggregation method
-

Q21. Which function is used to return Top N values in a table?

- A) FILTER
 - B) RANKX
 - C) TOPN**
 - D) ORDERBYCOLUMNS
-

Q22. To get Top 5 products by Sales, you would use:

- A) TOPN(5, Products, [Total Sales], DESC)**
 - B) RANKX(Products, [Total Sales], , DESC, 5)
 - C) MAXX(Products, [Total Sales])
 - D) FILTER(Products, [Total Sales] >= 5)
-

Q23. Which function modifies the filter context?

- A) CALCULATE**
 - B) SUMX
 - C) VALUES
 - D) RELATED
-

Q24. EARLIER() is mostly used in:

- A) Time intelligence functions
 - B) Row context inside calculated columns**
 - C) Measures with multiple filters
 - D) Concatenating strings
-

Q25. Which is NOT a valid DAX time intelligence function?

- A) NEXTDAY
- B) DATESINPERIOD
- C) FIRSTDATE
- D) TO_DATE**

DATA VISUALIZATION & INTERACTIVITY

Q1. Which chart is best to show trends over time in Power BI?

- A) Pie Chart
 - B) Line Chart**
 - C) Tree Map
 - D) Gauge
-

Q2. Which visualization is best for showing contribution of categories to a whole?

- A) Pie/Donut Chart**
 - B) Line Chart
 - C) Matrix Table
 - D) Scatter Plot
-

Q3. What does a KPI visual in Power BI primarily show?

- A) Hierarchies
 - B) Actual vs Target values**
 - C) Data distribution
 - D) Geographic locations
-

Q4. Which visual is best for showing hierarchical data?

- A) Tree Map**
 - B) Pie Chart
 - C) Matrix
 - D) Line Chart
-

Q5. Which visual is best suited for comparing performance across multiple categories side by side?

- A) Column Chart**
 - B) Gauge
 - C) Pie Chart
 - D) Scatter Plot
-

Q6. A scatter plot in Power BI is used to show:

- A) Distribution of categorical data
 - B) Relationship between two numerical variables**
 - C) Part-to-whole relationships
 - D) Ranking of categories
-

Q7. Which visualization is best to display actual vs forecasted values?

- A) Line Chart with multiple series**
 - B) Tree Map
 - C) KPI
 - D) Matrix
-

Q8. In a Power BI matrix visual, you can expand and collapse levels using:

- A) Filters
 - B) Hierarchies**
 - C) Slicers
 - D) Drillthrough
-

Q9. Which feature allows clicking on a chart element (like a bar) to filter data in other visuals?

- A) Drillthrough
- B) Cross-filtering / Cross-highlighting**

- C) Tooltips
 - D) Bookmarks
-

Q10. Slicers in Power BI are mainly used for:

- A) Data modeling
 - B) Adding interactivity with filters**
 - C) Creating measures
 - D) Formatting visuals
-

Q11. Which type of slicer allows selecting values from a continuous range?

- A) List slicer
 - B) Dropdown slicer
 - C) Between slicer (range slider)**
 - D) Hierarchy slicer
-

Q12. Drill-through in Power BI allows:

- A) Navigating to a detailed report page filtered for the selected item**
 - B) Exporting data into Excel
 - C) Adding interactivity to slicers
 - D) Combining multiple queries
-

Q13. What are bookmarks used for in Power BI?

- A) Saving filtered states and navigation views**
 - B) Creating DAX calculations
 - C) Improving refresh performance
 - D) Publishing reports
-

Q14. Which feature helps build navigation buttons between report pages?

- A) Drillthrough
 - B) Bookmarks with Buttons**
 - C) Tooltip Pages
 - D) Slicers
-

Q15. Conditional formatting in visuals can be applied to:

- A) Tables and Matrix
 - B) Bar and Column Charts (Data Bars, Colors)
 - C) KPI cards
 - D) All of the above**
-

Q16. Which visualization is best to compare performance against a set goal?

- A) Scatter Plot
 - B) Gauge**
 - C) Pie Chart
 - D) Matrix
-

Q17. What is the difference between Table and Matrix visual in Power BI?

- A) Table supports hierarchies; Matrix does not
 - B) Matrix supports hierarchies & drill-down; Table is flat**
 - C) Both are identical
 - D) Matrix is only used for time-series data
-

Q18. A combo chart in Power BI typically combines:

- A) Bar and Pie chart
 - B) Column and Line chart**
 - C) Scatter and Gauge chart
 - D) Map and Tree map
-

Q19. Which visual best represents geographic data in Power BI?

- A) Scatter Plot
 - B) Map Visual**
 - C) KPI
 - D) Table
-

Q20. Tooltips in Power BI are used for:

- A) Exporting data
 - B) Showing additional data on hover**
 - C) Creating slicers
 - D) Formatting visuals
-

Q21. Which feature lets you apply multiple filters across pages consistently?

- A) Report-level filters**
 - B) Drillthrough
 - C) Cross-filtering
 - D) Slicers only
-

Q22. Drill-down in a chart is enabled when:

- A) A hierarchy exists in the data fields**
 - B) Filters are disabled
 - C) Data is unpivoted
 - D) Bookmarks are applied
-

Q23. Which is the best visual to show top 10 products by sales?

- A) Tree Map**
- B) Stacked Column Chart
- C) Line Chart
- D) Gauge

Q24. Which chart type would you use to show correlation between revenue and customer age?

- A) Scatter Plot**
 - B) Gauge
 - C) Donut Chart
 - D) Table
-

Q25. Which interactivity feature allows users to customize the report view without editing it?

- A) Personal Bookmarks**
 - B) Drillthrough Pages
 - C) Model Relationships
 - D) Append Queries
-

Q26. Which Power BI feature allows showing different visuals based on user selection (e.g., Sales vs Profit)?

- A) Drillthrough
 - B) Hierarchies
 - C) Bookmarks with Buttons (Toggle)**
 - D) Tooltips
-

Q27. What is the main advantage of using a decomposition tree visual in Power BI?

- A) Displays trend over time
 - B) Compares actual vs target
 - C) Breaks down metrics into contributing factors interactively**
 - D) Shows part-to-whole relationships
-

Q28. Which formatting option allows you to show values directly inside a bar or column chart?

- A) Conditional Formatting
 - B) Tooltip
 - C) Data Labels**
 - D) Cross-filtering
-

Q29. What is the purpose of the “Sync slicers” feature in Power BI?

- A) Create drillthrough filters
 - B) Apply the same slicer selection across multiple report pages**
 - C) Combine two different slicers into one
 - D) Export slicer data to Excel
-

Q30. Which interactive feature helps highlight data in visuals without filtering other visuals?

- A) Drillthrough
- B) Cross-highlighting**
- C) Report-level filters
- D) Bookmarks

REPORTING, DASHBOARDS & AI INTEGRATION

Q1. In Power BI, which feature allows you to save a specific state of a report page (filters, visuals) for easy navigation?

- A) Drillthrough
 - B) Bookmarks**
 - C) Hierarchies
 - D) Tooltips
-

Q2. Which feature allows the creation of interactive buttons for navigation within a Power BI report?

- A) Filters
 - B) Action Buttons with Bookmarks**
 - C) Relationships
 - D) Drillthrough
-

Q3. What is the primary purpose of a dashboard in Power BI Service?

- A) To build relationships
 - B) To clean and transform data
 - C) To pin visuals from different reports into a single view**
 - D) To create measures and columns
-

Q4. Which Power BI Service feature allows real-time updates in dashboards from streaming datasets?

- A) Report publishing
- B) Import Mode
- C) Push/Streaming Datasets**
- D) Drillthrough

Q5. Which type of filter applies to the entire report across all pages?

- A) Visual-level filter
 - B) Page-level filter
 - C) Report-level filter**
 - D) Slicer filter
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Q6. Which feature allows you to provide a guided story-like experience in Power BI reports?

- A) Drillthrough
 - B) Bookmarks with Selection Pane**
 - C) Cross-highlighting
 - D) Sync slicers
-

Q7. What is the purpose of KPI visuals in Power BI?

- A) Show trend comparisons
 - B) Track progress against a defined target**
 - C) Display hierarchy-based data
 - D) Break data into components
-

Q8. Which Power BI feature allows sharing dashboards securely with colleagues?

- A) Export to Excel
 - B) Report Filters
 - C) Power BI Service Sharing**
 - D) Drillthrough
-

Q9. Which type of refresh ensures the latest data is shown in a published dashboard?

- A) Manual refresh

- B) Import refresh
 - C) Scheduled refresh**
 - D) Reset refresh
-

Q10. Which view in Power BI Desktop is used to design the layout of report visuals?

- A) Data View
 - B) Model View
 - C) Report View**
 - D) Dashboard View
-

Q11. Which feature allows you to drill into a specific data point in a different report page?

- A) Bookmarks
 - B) Filters
 - C) Drillthrough**
 - D) Hierarchies
-

Q12. What is the main advantage of using slicers in dashboards?

- A) To export visuals
 - B) To filter data interactively**
 - C) To create new tables
 - D) To combine datasets
-

Q13. Which Power BI feature helps in aligning and arranging visuals neatly on a report page?

- A) Data View
 - B) Drillthrough
 - C) Snap-to-Grid & Alignment Guides**
 - D) Sync slicers
-

Q14. What does “Pinning” a visual mean in Power BI?

- A) Exporting it to Excel
 - B) Adding it to a dashboard from a report**
 - C) Applying a filter
 - D) Saving it as a template
-

Q15. Which file format is used to save and share Power BI reports?

- A) .xls
 - B) .csv
 - C) .pbix**
 - D) .docx
-
-

AI INTEGRATION

Q16. Which AI visual in Power BI explains the key drivers behind a metric?

- A) Q&A
 - B) Key Influencers**
 - C) Decomposition Tree
 - D) Forecasting
-

Q17. Which Power BI AI feature allows natural language queries like “Show sales by region”?

- A) Forecasting
 - B) Q&A Visual**
 - C) Key Influencers
 - D) Tooltips
-

Q18. Which visual allows breaking down data step by step to analyze contributions?

- A) Gauge
 - B) KPI
 - C) Decomposition Tree**
 - D) Donut
-

Q19. What does the AI feature “Analyze” (right-click on a visual) do in Power BI?

- A) Export data
 - B) Automatically detect explanations like “Explain the increase”**
 - C) Refresh datasets
 - D) Create new bookmarks
-

Q20. Which AI feature can project future values based on historical data in line charts?

- A) Key Influencers
 - B) Decomposition Tree
 - C) Forecasting**
 - D) Smart Narratives
-

Q21. Smart Narratives in Power BI are used to:

- A) Clean data
 - B) Automatically generate textual summaries of visuals**
 - C) Build relationships
 - D) Add bookmarks
-

Q22. Which feature allows users to type questions and get visuals instantly?

- A) Smart Narratives
 - B) Key Influencers
 - C) Q&A**
 - D) Drillthrough
-

Q23. Which AI integration provides explanations like “This category contributed the most to variance”?

- A) Tooltips
 - B) Analyze (AI Insights)**
 - C) Cross-highlighting
 - D) Bookmarks
-

Q24. Which AI feature suggests the most significant factors influencing a selected outcome?

- A) Forecasting
 - B) Key Influencers**
 - C) Decomposition Tree
 - D) Q&A
-

Q25. Which AI feature is most useful for executives who want plain language insights directly in reports?

- A) Q&A
 - B) Smart Narratives**
 - C) Bookmarks
 - D) KPI
-

Q26. In Power BI, AutoML in Premium capacity allows users to:

- A) Create relationships
 - B) Train and apply machine learning models**
 - C) Create dashboards
 - D) Build slicers
-

Q27. AI Insights in Power Query provides:

- A) DAX formulas

B) Prebuilt cognitive services like text and sentiment analysis

C) Visualization templates

D) Dashboard alignment

Q28. Which feature allows combining Azure Cognitive Services with Power BI data for AI analysis?

A) Smart Narratives

B) Key Influencers

C) AI Insights

D) Drillthrough

Q29. Which Power BI AI feature is best for “what-if” analysis of contributing factors?

A) Q&A

B) Decomposition Tree

C) KPI

D) Filters

Q30. Which AI-driven visual in Power BI is designed to answer “Why did this happen?” questions?

A) Forecasting

B) Key Influencers

C) Q&A

D) Smart Narratives